# ADVANCE Disadoption Study v5

Behavioural correlates of e-cigarette adoption and disadoption (W1-W10)

A. Olivera, T. Valente, K. Miljkovic, Y. Cao

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# 1. Introduction

v5 keeps the v4b panel (full **W1-W10** ADVANCE panel — 10 semester waves, 4,437 students, 042326 release) and switches the time control from `wave fixed effects + cohort dummy` to **`grade-semester fixed effects (gs_fe) + cohort dummy`**. Grade-semester is derived from `(cohort, wave)` on the standard ADVANCE timeline (1 = fall 9th, 2 = spring 9th, ..., 8 = spring 12th); it answers the prevalent reviewer concern that the time control should track *developmental* position within high school rather than calendar time. The cohort dummy is retained (cohorts experience the same grade-semester at different calendar times). Six regression families are reported:

- **§5 Main**: the v4 spec ($E_D$ = peer share who flipped $1 \to 0$ between $w-2$ and $w-1$).
- **§6 Alt** $E_D$: alternative operationalisation $E_D = E^{\max}_{i,1..w-1} - E_{\text{users},i,w-1}$ (cumulative peak − current exposure).
- **§7 No** $E_D$: refit §5 outcomes (Adopters, A, B, C) without the $E_D$ predictor.
- **§8 Model C window sensitivity**: $W = 1$, $W \leq 2$, $W \leq 3$ for the cyclic disadoption definition.
- **§9 Sensitivity (a)**: $1 \to 0$ with NA-only future counted as Stable (A).
- **§10 Sensitivity (b)**: event identification using observed-wave jumps rather than consecutive-calendar pairs.

All regression cells report **odds ratios** $\exp(\hat{\beta})$ with the cluster-robust (or `glmer`) p-value in parentheses. **Bold** marks $p < 0.05$.

# 2. Data

## 2.1 Sources and panel construction

- **W1-W8**: `data/advance/Data/ADVANCE_W1-W8_Data_Complete_042326.xlsx` (4,437 students).
- **W9-W10 HS**: `data/advance/Data/ADVANCE_W9-W10_HS_Data_Complete_042326.xlsx` (1,060 students, strict subset of W1-W8).
- **Edges**: regenerated from the 042326 XLSX into `data/advance/Cleaned-Data-042326/wNedges_clean.csv` for $N = 1..10$. Uniform rule: keep edge $(i, j)$ iff $j \neq i$, $j$ is in the panel, and $j$ responded to wave $w$. W1 edges adopt the legacy `Cleaned-Data/w1edges_clean.csv` after applying the same hygiene (the W1 XLSX stores friend cells as REDCap-internal codes without a record_id map).

Per-wave non-NA `past_6mo_use_3` counts: W1 = 851, W2 = 2,235, W3 = 3,768, W4 = 3,907, W5 = 3,833, W6 = 3,645, W7 = 3,378, W8 = 3,048, **W9 = 974, W10 = 969**.

Per-wave edge counts (clean): W1 = 1,466, W2 = 4,344, W3 = 9,681, W4 = 10,133, W5 = 9,987, W6 = 8,850, W7 = 8,120, W8 = 6,879, **W9 = 2,665, W10 = 2,511**. Self-loops dropped in all waves; reciprocity 50–55%.

#### 2.1.1 Edge similarity vs the legacy `Cleaned-Data/`

To verify that the 042326 reconstruction reproduces the legacy edge set, we compute Jaccard similarity per wave and the share of legacy edges retained ("% old kept") and reciprocally:

| Wave | n_legacy | n_v4b | both | Jaccard % | % old kept | % new in old |
|---|---|---|---|---|---|---|
| 1 | 1,474 | 1,466 | 1,466 | 99.5 | 99.5 | 100.0 |
| 2 | 4,380 | 4,344 | 4,342 | 99.1 | 99.1 | 100.0 |
| 3 | 9,840 | 9,681 | 9,649 | 97.7 | 98.1 | 99.7 |
| 4 | 10,156 | 10,133 | 10,073 | 98.8 | 99.2 | 99.4 |
| 5 | 9,928 | 9,987 | 9,901 | 98.9 | 99.7 | 99.1 |
| 6 | 8,840 | 8,850 | 8,767 | 98.6 | 99.2 | 99.1 |
| 7 | 8,134 | 8,120 | 8,083 | 99.3 | 99.4 | 99.5 |
| 8 | 6,871 | 6,879 | 6,773 | 97.5 | 98.6 | 98.5 |

Jaccard  97% in W1–W8. In W1–W8,  98% of legacy edges are present in v4b, and 100% of v4b edges in W1, W2, W9, W10 were already in the legacy set.

### 2.2 Q-restriction

Eligibility per $Q \in \{5, 6, 7, 8\}$ requires the student to have at least $Q$ **consecutive** observed waves of `past_6mo_use_3`. Adding W9–W10 dramatically expands the high-$Q$ samples:

| Q (consecutive) | v4 (W1-W8) | v4b (W1-W10) |
|---|---|---|
| 8 | 371 | **1,040** |
| 7 | 1,228 | **1,961** |
| 6 | 2,453 | **2,499** |
| 5 | 2,972 | **3,007** |

Network alters used to compute $E_{\text{users}}$ and $E_D$ are NOT restricted by Q.

**Friendship-nomination degree distribution (pooled W1-W10).** Each ego nominates up to seven friends per wave; out-degree is the count of valid alters (alter in the panel and responding at $w$); in-degree is how many other egos name a given student.

Out-degree is mechanically capped near 7 (questionnaire limit); in-degree has a long right tail (a few students are frequently nominated). Both distributions are right-skewed but well populated.

**Eligible vs effective**: the Q-row above is the count of *eligible* students. The regression in each column then drops rows with NA on any predictor (complete-case). The effective `N Students` reported per column is therefore smaller than the Q-row count. The Adopters column suffers the smallest cut (most predictors are filled at every observed wave); the A / B / C columns work on the much smaller ever-user subset (each column's `N Students` reflects that). E.g., at $Q = 8$ we have 1,040 eligible students; the Adopters regression uses 925 (after complete-case dropping); the A regression uses 83 (ever-users with valid predictors).

### 2.3 Parent education — scale harmonisation and per-student LOCF

Parent education is the only covariate that required non-trivial cross-wave handling: the questionnaire scale itself **changed mid-panel**, from a 7-level scale in W1–W6 to a 9-level scale in W7–W10. We harmonise to the legacy 7-level scale and then carry forward each student's most recent observation to fill the gaps.

**Step 1 — Raw column.** Each wave has the column `w{w}_dem_high_par_edu` (highest parental education level reported by the student). It is read selectively by `R/01-advance-panel.R` (line 30) into the long panel as `par_edu_raw` (line 139 in the wide→long step).

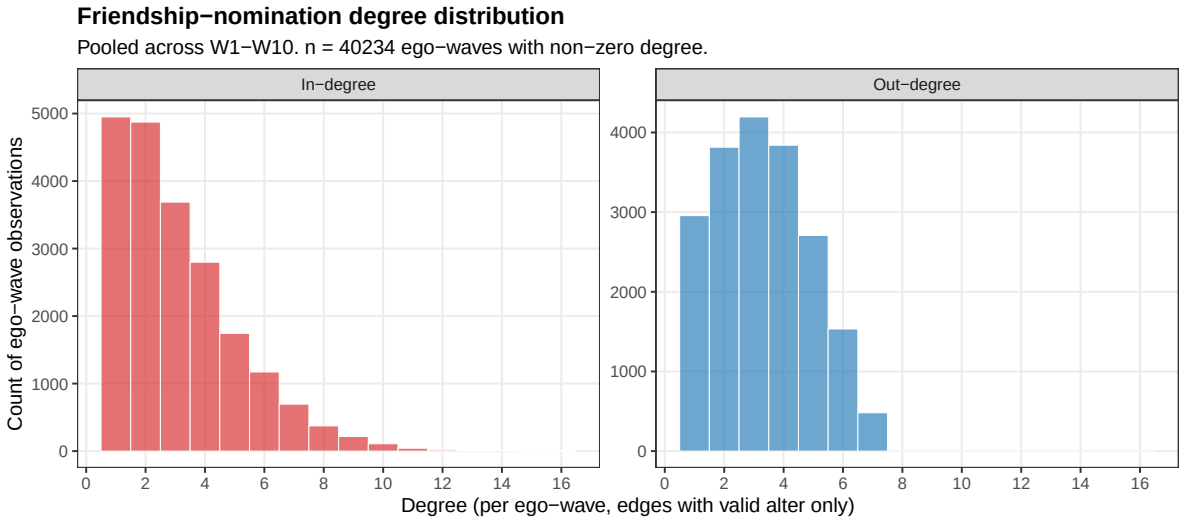



Figure 1: Out-degree and in-degree distributions across all valid (ego, wave) cells with non-zero degree.

**Step 2 — Scale change at W7.** From W7 onward the questionnaire was extended to 9 categories (additional fine-grained options between "some college" and "bachelor's"). Codes 1–4 are stable across versions; the legacy "4" category was split into three (4, 5, 6) in the new scale; legacy 5/6/7 map to new 7/8/9. We collapse the new scale to the legacy scale via the deterministic remap below (`R/01-advance-panel.R:103-114`):

| New scale (W7-W10) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| Legacy scale (W1-W6) | 1 | 2 | 3 | 4 | **4** | **4** | 5 | 6 | 7 |

The collapse of new codes 4/5/6 onto legacy 4 is information-losing (three categories become one) but is the only mapping that produces a unified ordinal scale across all ten waves. After remapping, `par_edu_raw` is bounded in $\{1, \ldots, 7\}$ at every wave.

**Step 3 — Per-student last-observation-carried-forward (LOCF).** Because students do not refresh their parental-education report every wave, many person-waves have `par_edu_raw = NA` between two observed values. We treat parent education as **slowly-varying (or effectively time-invariant) within student** and carry the most recent observed value forward to fill subsequent NAs. The LOCF function in `R/01-advance-panel.R:191-198` is applied per `record_id`, producing the final regression-ready `par_edu` column. LOCF only overwrites NAs; observed values are never replaced by earlier observations.

**Resulting time profile.** This pipeline produces a small but real cross-wave shift in the panel mean: at gs = 2 (spring 9th, mostly W2 observations on the legacy 7-level scale) the mean `par_edu` is 5.07; at gs = 8 (spring 12th, mostly W8 observations on the remapped 9-level scale) it falls to 3.82 because the three new-scale categories 4/5/6 all collapse to legacy 4, pulling the upper tail down. **This is a measurement artifact, not a behavioural shift in family socioeconomic status.** Within-student variation in `par_edu` over time is small (most students report the same value across waves) and the regression specifications include grade-semester FE that absorb mean shifts of this kind, so the artefact does not contaminate the per-predictor ORs. The artefact would matter if we treated `par_edu` as a *time-varying* shock — we don't.

# 3. Event definitions

For each student we define, on consecutive observed waves:

- **Adopters**: first $0 \to 1$ transition. One event per person.

- **A — Stable**: $1 \to 0$ with **no future return to 1** in any later observed wave. One event per person. Indeterminates ($1 \to 0$ with NA-only future) are dropped from A in §5/§6/§8/§10; counted as A in §9.
- **B — Experimental**: first $1 \to 0$ (any). One event per person.
- **C — Unstable** (window $W$): $1 \to 0$ followed by 1 within the next $W$ observed waves. Multiple events per person possible. §5/§6/§9/§10 use $W = 1$; §8 reports $W = 1, 2, 3$.

§10 walks through observed waves regardless of calendar gaps (instead of consecutive-calendar pairs).

**Per-wave degree and using-friend count distributions.** For every wave $w \in \{1, \ldots, 10\}$ the figure below overlays *two* distributions: out-degree (blue bars, "how many ego-waves nominate $k$ friends?") and the count of *using* friends per ego (red bars, "how many ego-waves have $k$ friends who currently use ecig?"). The "count of using friends" is $k_{\text{users}} = E_{\text{users}} \times$ out_degree (rounded to integer): the number of alters of the ego who currently use ecig at that wave.

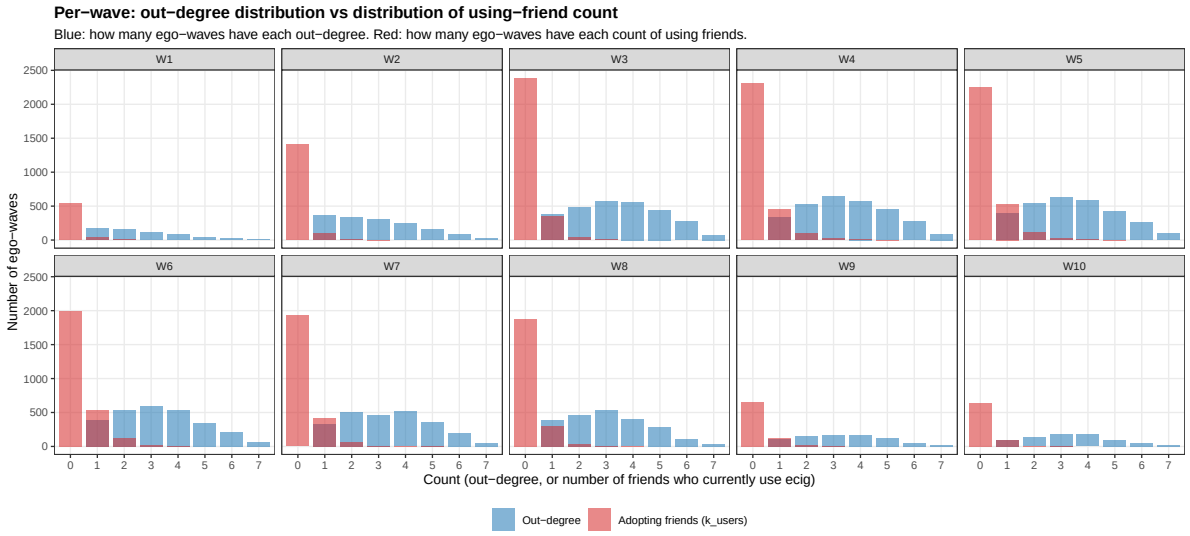



Figure 2: Per-wave: blue = distribution of out-degree (count of friends nominated); red = distribution of $k_{\text{users}}$ (count of friends who currently use ecig). Both at integer support; bars at the same x-value are overlaid with alpha = 0.55 so overlap is visible.

Two patterns are visible: (i) out-degree is centered around 3-4 friends in every wave (questionnaire cap = 7), confirming a stable nomination structure; (ii) the using-friend distribution is heavily concentrated at $k_{\text{users}} = 0$ in early waves but the right tail thickens steadily from W3 onward. By W6-W8 a substantial slice of egos has 1-2 using friends; very few egos ever exceed 3 using friends.

# 4. Methods

For each regression, the outcome at person-wave $(i, w)$ is binary; the risk-set is the corresponding panel:

- **Adopters / A / B**: GLM logistic with **grade-semester fixed effects** (`gs_fe`, 8 levels = 7 dummies, reference = gs=1) and cluster-robust SE by `record_id` (`sandwich::vcovCL`).
- **C**: `lme4::glmer(... + (1 \mid \text{record\_id}))` with `gs_fe`, since C admits multiple events per person; we report ICC $\rho = \sigma_u^2/(\sigma_u^2 + \pi^2/3)$.

**Grade-semester encoding**. `gs_fe` is derived from `(cohort, wave)` on the ADVANCE timeline (class of 2024: W1 = gs 1 (fall 9th), W2 = gs 2 (spring 9th), …, W8 = gs 8 (spring 12th); W9-W10 are post-HS, set to NA and excluded. Class of 2025: W3 = gs 1, …, W10 = gs 8). Each grade-semester spans roughly six months. The first observed wave for each cohort produces no at-risk rows because the lag is undefined; this leaves gs=1 with zero at-risk rows and the reference category for `gs_fe` is therefore gs=2 (spring 9th) in

practice. Grade-repeaters ( 1–3% of HS students nationally) introduce a small mislabeling bound of ~1% of person-waves; the cohort dummy partially absorbs the systematic component.

**Predictors (13)**: cohort (2025 vs 2024), female, sexual minority, parent education, asian, hispanic/latine, MDD (RCADS Mean), GAD (RCADS Mean), out-degree, in-degree, perceived friend use $(w-1)$, network exposure to users $(E_{\text{users}}, w-1)$, network exposure to dis-adopters $(E_D, w-1)$. With W9-W10 in the panel, both cohorts are retained at all $Q$ levels: cohort 2024 students with $\geq Q$ consecutive observed waves and cohort 2025 students whose $w \geq 3$ entry leaves them with $\geq Q$ observed waves through W10.

**Note on ESE**: the Early Smoking Experience composites (`ESE_Pos_no9_Mean`, `ESE_Neg_no510_Mean`) are **not used as predictors** in v4b — only ~21% of students have any ESE response, and the missingness is concentrated outside the at-risk-for-disadoption sub-population. Including ESE forces complete-case dropping that selects toward users and biases the Adopters and A/B/C samples. Sensitivity work on the user-subset is deferred to a later iteration.

§6 substitutes $E_D = E^{\max}_{i,1..w-1} - E_{\text{users},i,w-1}$ for the §5 definition.

**Raw XLSX inputs.** All variables in v4b are constructed from two release files: `ADVANCE_W1-W8_Data_Complete_042326.xls` (4,437 students; W1-W8) and `ADVANCE_W9-W10_HS_Data_Complete_042326.xlsx` (1,060 students; W9-W10, strict subset of W1-W8). Per wave $w \in \{1, \ldots, 10\}$ we consume **only** the columns below; everything else in the wide release is ignored.

| Raw XLSX column (per wave $w$) | Long-panel field | Used to build |
|---|---|---|
| `record_id` (no wave prefix) | `record_id` | row anchor |
| `w{w}_schoolid` | `schoolid` | community FE; cohort assignment via first non-NA school |
| `w{w}_past_6mo_use_3` | `ecig` | **outcome** ($1 \to 0$ = disadoption event); `ecig` at $w-1$ defines who is at risk |
| `w{w}_past_6mo_use_2` | `cig` | retained for reference; not a v4b predictor |
| `w{w}_dem_gender` | `dem_gender` | `female = 1[\text{dem\_gender}=0]`; code `3` ("prefer not to disclose") → NA |
| `w{w}_dem_sexuality` | `dem_sexuality` | `sex_minority = 1[\text{dem\_sexuality} \neq 1]`; code `10` → NA |
| `w{w}_dem_high_par_edu` | `par_edu_raw` → `par_edu` | LOCF per student; W7-W10 1–9 scale remapped to W1-W6 1–7 scale |
| `w{w}_race` | `race` | `asian = 1[\text{race}=2]`; W4/W5 code `8` ("declined") → NA |
| `w{w}_eth` | `eth` | `hispanic = 1[\text{eth}=1]` |
| `w{w}_rcads_mdd_mean` | `mdd` | MDD predictor (RCADS depression mean) |
| `w{w}_rcads_gad_mean` | `gad` | GAD predictor (RCADS anxiety mean) |
| `w{w}_ese_ecig_pos_no9_mean` | `ese_pos` | ESE positive composite — *read but not used in v4b regressions* |
| `w{w}_ese_ecig_neg_no510_mean` | `ese_neg` | ESE negative composite — *read but not used in v4b regressions* |
| `w{w}_friends_use_ecig` | `friends_use_ecig` | **PFU** (`Perceived Friend Use`); used at $w-1$ as `friends_use_ecig_lag`. Code `6` ("Not sure") → NA |

| Raw XLSX column (per wave $w$) | Long-panel field | Used to build |
|---|---|---|
| `w{w}_friend{1..7}` (W1) / `w{w}_friend{1..7}_{schoolid}` (W2-W10) | edges | friendship nominations → `data/advance/Cleaned-Data-042326/wNedges_c` out-degree, in-degree, $E_{\text{users}}$, $E_D$ |

The cohort dummy is derived from the first non-NA `schoolid` per student (schools 101–108, 112–114 → 2024; 201, 212–214 → 2025). W1 schools are stored as `1..5` and remapped to `101..105`. W9/W10 `schoolid = 999` (transferred-out) is treated as NA.

**Effective sample size by panel × Q.** Each cell is *students / events*. "Full" = at-risk panel after the Q-restriction; "CC" = panel after `complete.cases` on the 13 predictors (= what each regression actually fits). Loss to complete-cases is mainly driven by $E_{\text{users}}$ / $E_D$ (~24% NA at $Q = 8$, both depend on alters' response at $w - 1$), `asian` (~20%), and the mental-health and PFU items (~17% each).

| Q | Adopt full | Adopt CC | A full | A CC | B full | B CC | C full | C CC |
|---|---|---|---|---|---|---|---|---|
| 8 | 1029 / 162 | 925 / 89 | 139 / 96 | 83 / 52 | 158 / 142 | 91 / 70 | 139 / 17 | 83 / 7 |
| 7 | 1930 / 346 | 1711 / 193 | 299 / 189 | 161 / 89 | 346 / 293 | 182 / 129 | 299 / 43 | 161 / 15 |
| 6 | 2441 / 449 | 2077 / 232 | 399 / 244 | 206 / 108 | 461 / 384 | 233 / 159 | 399 / 66 | 206 / 21 |
| 5 | 2919 / 551 | 2404 / 267 | 493 / 288 | 237 / 124 | 568 / 467 | 271 / 185 | 493 / 81 | 237 / 29 |

The two figures below visualise the same N table. Each bar is **one rectangle per (panel, Q)** with `N students` in low-alpha shading and `N events` overlaid in high-alpha shading of the same colour, so the gap between the two values is the count of *at-risk students who never disadopt at the corresponding Q level.*

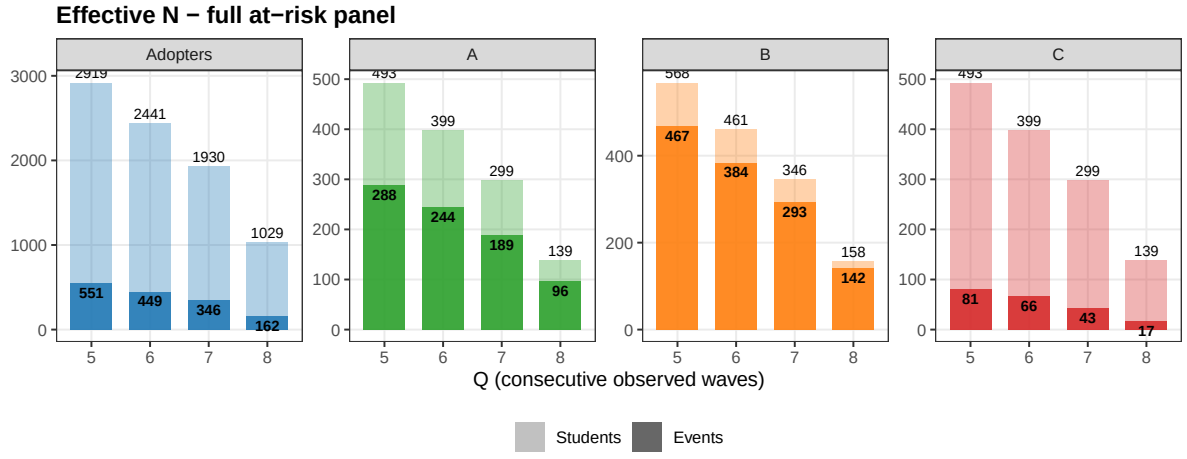



Figure 3: Effective N — full at-risk panel (before complete-cases). Each bar shows total at-risk students with events overlaid.

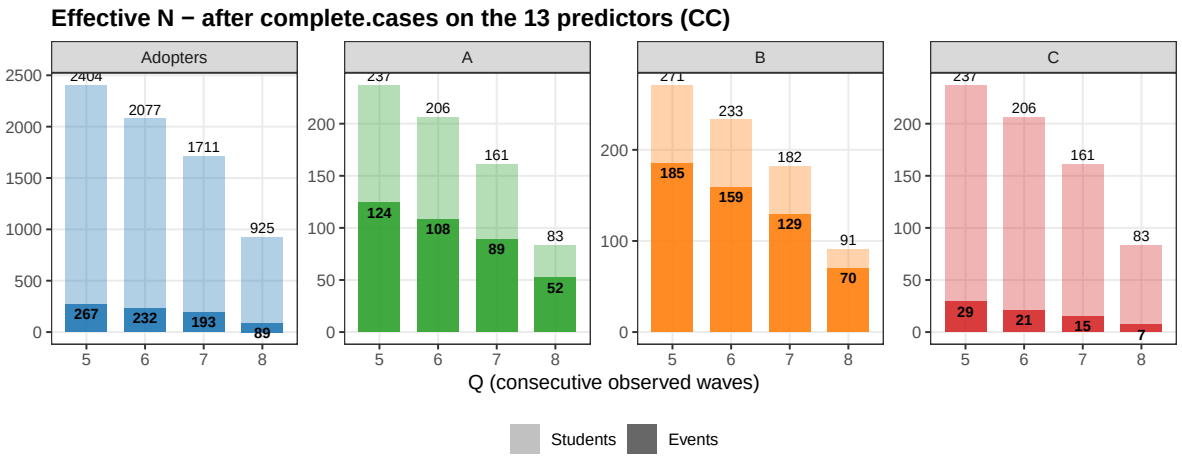



Figure 4: Effective N — after complete.cases on the 13 predictors (= what each regression actually fits).

# 5. Main results (E_D = peer-flipped 1→0; C window = 1)

OR (p-value). Bold = $p < 0.05$. $E_D$ = peer share who flipped $1 \to 0$ between $w-2$ and $w-1$. *Model C uses a person random intercept; $\rho = 0.000$ across all fits, so conditional and marginal ORs coincide empirically.*

## 5.1 Q = 8

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 1.138 (0.584) | 1.771 (0.424) | 0.876 (0.839) | 2.248 (0.613) |
| Female | 1.635 (0.065) | **0.336 (0.044)** | 0.399 (0.109) | 0.855 (0.926) |
| Sexual Minority | 1.546 (0.089) | 1.279 (0.687) | 1.733 (0.324) | **21.702 (0.040)** |
| Parent Ed. | 0.999 (0.992) | 1.158 (0.496) | 0.933 (0.800) | 0.327 (0.175) |
| Asian | 0.577 (0.056) | 1.188 (0.800) | 0.666 (0.480) | 0.264 (0.299) |
| Hispanic/Latine | 1.031 (0.917) | 0.724 (0.668) | 0.584 (0.389) | 0.009 (0.059) |
| MDD (Major Depressive S.) | 1.223 (0.421) | 0.502 (0.140) | **0.361 (0.018)** | **0.028 (0.019)** |
| GAD (Generalized Anxiety Dis.) | 0.877 (0.563) | 2.470 (0.079) | 1.907 (0.184) | 10.328 (0.129) |
| Out-degree | **0.829 (0.013)** | 1.356 (0.109) | 1.031 (0.875) | 0.683 (0.485) |
| In-degree | **1.103 (0.045)** | 1.124 (0.406) | 1.020 (0.878) | 0.556 (0.268) |
| Perceived Friend Use | **1.462 (0.000)** | 0.904 (0.657) | 0.941 (0.719) | 1.843 (0.329) |
| Network Exposure Users | **3.726 (0.034)** | **0.021 (0.018)** | **0.068 (0.010)** | 0.080 (0.447) |
| Network Exposure Dis-adopters | 3.023 (0.262) | 1.211 (0.892) | 0.482 (0.629) | 0.059 (0.691) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 925 | 83 | 91 | 83 |
| N Events | 89 | 52 | 70 | 7 |

## 5.2 Q = 7

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 0.860 (0.367) | 1.705 (0.247) | 1.171 (0.715) | 0.535 (0.444) |
| Female | 1.345 (0.103) | 0.760 (0.482) | 1.073 (0.850) | 3.125 (0.163) |
| Sexual Minority | **1.461 (0.028)** | 0.734 (0.406) | 0.859 (0.670) | 1.165 (0.817) |

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Parent Ed. | 0.974 (0.662) | 1.014 (0.922) | 0.863 (0.346) | 0.584 (0.068) |
| Asian | **0.627 (0.017)** | 0.979 (0.963) | 1.036 (0.927) | 0.401 (0.237) |
| Hispanic/Latine | 1.070 (0.729) | 0.882 (0.781) | 0.691 (0.314) | 0.274 (0.104) |
| MDD (Major Depressive S.) | 1.171 (0.369) | 0.741 (0.380) | **0.454 (0.007)** | **0.217 (0.025)** |
| GAD (Generalized Anxiety Dis.) | 0.863 (0.346) | 1.451 (0.264) | 1.181 (0.573) | 1.305 (0.676) |
| Out-degree | 0.913 (0.075) | 1.184 (0.204) | 0.937 (0.607) | 0.915 (0.705) |
| In-degree | **1.100 (0.007)** | 1.103 (0.265) | 1.128 (0.111) | 1.075 (0.654) |
| Perceived Friend Use | **1.474 (0.000)** | **0.710 (0.013)** | **0.698 (0.001)** | 0.748 (0.245) |
| Network Exposure Users | **5.427 (0.000)** | **0.135 (0.016)** | 0.294 (0.058) | 1.184 (0.903) |
| Network Exposure Dis-adopters | 2.650 (0.200) | 0.164 (0.169) | 0.785 (0.767) | 0.704 (0.896) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 1711 | 161 | 182 | 161 |
| N Events | 193 | 89 | 129 | 15 |

## 5.3 Q = 6

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 0.827 (0.226) | 1.851 (0.133) | 1.389 (0.392) | 0.431 (0.262) |
| Female | 1.303 (0.095) | 0.596 (0.130) | 0.826 (0.560) | 2.669 (0.177) |
| Sexual Minority | **1.384 (0.038)** | 0.790 (0.456) | 1.078 (0.803) | 2.152 (0.160) |
| Parent Ed. | 0.968 (0.554) | 0.987 (0.911) | 0.950 (0.697) | 0.792 (0.301) |
| Asian | **0.622 (0.008)** | 1.089 (0.827) | 1.080 (0.833) | 0.468 (0.252) |
| Hispanic/Latine | 1.126 (0.498) | 0.863 (0.690) | 0.866 (0.655) | 0.436 (0.183) |
| MDD (Major Depressive S.) | 1.114 (0.500) | 0.728 (0.306) | **0.516 (0.017)** | 0.361 (0.068) |
| GAD (Generalized Anxiety Dis.) | 0.891 (0.419) | 1.241 (0.453) | 1.145 (0.612) | 1.597 (0.381) |
| Out-degree | 0.936 (0.153) | 1.168 (0.177) | 1.009 (0.930) | 0.996 (0.985) |
| In-degree | **1.089 (0.008)** | 1.052 (0.499) | 1.091 (0.175) | 1.106 (0.439) |
| Perceived Friend Use | **1.506 (0.000)** | **0.700 (0.001)** | **0.703 (0.000)** | 0.764 (0.158) |
| Network Exposure Users | **5.493 (0.000)** | 0.324 (0.110) | 0.402 (0.117) | 1.038 (0.975) |
| Network Exposure Dis-adopters | 1.782 (0.423) | 0.374 (0.339) | 1.149 (0.860) | 0.702 (0.865) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 2077 | 206 | 233 | 206 |
| N Events | 232 | 108 | 159 | 21 |

## 5.4 Q = 5

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 0.836 (0.213) | 1.786 (0.116) | 1.379 (0.337) | 0.557 (0.301) |
| Female | **1.431 (0.016)** | 0.547 (0.062) | 0.732 (0.317) | 2.198 (0.193) |
| Sexual Minority | **1.435 (0.013)** | 0.877 (0.639) | 1.060 (0.826) | 1.596 (0.295) |
| Parent Ed. | 0.948 (0.301) | 0.972 (0.811) | 0.935 (0.575) | 0.838 (0.339) |
| Asian | **0.678 (0.022)** | 1.133 (0.735) | 1.086 (0.809) | 0.424 (0.145) |
| Hispanic/Latine | 1.162 (0.362) | 0.858 (0.642) | 0.905 (0.732) | 0.533 (0.211) |
| MDD (Major Depressive S.) | 1.137 (0.375) | 0.795 (0.415) | **0.536 (0.015)** | **0.382 (0.045)** |

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| GAD (Generalized Anxiety Dis.) | 0.840 (0.190) | 1.181 (0.517) | 1.217 (0.422) | 1.501 (0.362) |
| Out-degree | 0.941 (0.164) | 1.135 (0.236) | 1.057 (0.562) | 1.038 (0.813) |
| In-degree | **1.081 (0.009)** | 1.045 (0.520) | 1.060 (0.322) | 1.047 (0.683) |
| Perceived Friend Use | **1.490 (0.000)** | **0.742 (0.002)** | **0.729 (0.000)** | 0.787 (0.110) |
| Network Exposure Users | **5.699 (0.000)** | 0.491 (0.253) | 0.701 (0.502) | 2.122 (0.359) |
| Network Exposure Dis-adopters | 2.926 (0.058) | 0.645 (0.651) | 1.184 (0.803) | 0.190 (0.430) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 2404 | 237 | 271 | 237 |
| N Events | 267 | 124 | 185 | 29 |

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# 6. Alternative E_D = E^max − E_current

Replaces "peer share who flipped 1→0 between $w-2$ and $w-1$" with $E_D = \max_{s \le w-1} E_{\text{users},i,s} - E_{\text{users},i,w-1}$.

## 6.1 Q = 8

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 1.135 (0.593) | 1.771 (0.418) | 0.948 (0.933) | 2.641 (0.538) |
| Female | 1.643 (0.063) | **0.344 (0.045)** | 0.438 (0.151) | 0.904 (0.954) |
| Sexual Minority | 1.534 (0.094) | 1.264 (0.699) | 1.675 (0.348) | **21.825 (0.043)** |
| Parent Ed. | 0.997 (0.972) | 1.179 (0.426) | 0.960 (0.881) | 0.306 (0.162) |
| Asian | 0.583 (0.062) | 1.189 (0.803) | 0.620 (0.408) | 0.282 (0.329) |
| Hispanic/Latine | 1.038 (0.899) | 0.757 (0.716) | 0.594 (0.419) | 0.010 (0.063) |
| MDD (Major Depressive S.) | 1.227 (0.416) | 0.498 (0.132) | **0.349 (0.012)** | **0.027 (0.022)** |
| GAD (Generalized Anxiety Dis.) | 0.879 (0.571) | 2.535 (0.080) | 1.942 (0.167) | 9.160 (0.139) |
| Out-degree | **0.830 (0.015)** | 1.367 (0.109) | 1.066 (0.749) | 0.648 (0.430) |
| In-degree | **1.106 (0.037)** | 1.120 (0.430) | 1.010 (0.936) | 0.541 (0.241) |
| Perceived Friend Use | **1.457 (0.000)** | 0.910 (0.674) | 0.942 (0.729) | 1.823 (0.329) |
| Network Exposure Users | **3.922 (0.029)** | **0.019 (0.020)** | **0.050 (0.006)** | 0.043 (0.332) |
| Network Exposure Dis-adopters | 1.734 (0.471) | 0.723 (0.818) | 0.295 (0.409) | 0.897 (0.983) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 925 | 83 | 91 | 83 |
| N Events | 89 | 52 | 70 | 7 |

## 6.2 Q = 7

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 0.857 (0.354) | 1.682 (0.255) | 1.207 (0.668) | 0.623 (0.568) |
| Female | 1.345 (0.105) | 0.834 (0.649) | 1.148 (0.718) | 3.393 (0.137) |
| Sexual Minority | **1.449 (0.031)** | 0.682 (0.297) | 0.836 (0.612) | 1.236 (0.753) |
| Parent Ed. | 0.972 (0.635) | 1.025 (0.856) | 0.878 (0.410) | 0.609 (0.091) |
| Asian | **0.637 (0.023)** | 0.913 (0.844) | 0.984 (0.966) | 0.349 (0.185) |
| Hispanic/Latine | 1.084 (0.683) | 0.826 (0.668) | 0.683 (0.302) | 0.260 (0.096) |
| MDD (Major Depressive S.) | 1.171 (0.369) | 0.736 (0.365) | **0.442 (0.006)** | **0.208 (0.023)** |

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| GAD (Generalized Anxiety Dis.) | 0.865 (0.351) | 1.427 (0.279) | 1.203 (0.529) | 1.361 (0.629) |
| Out-degree | 0.914 (0.078) | 1.183 (0.209) | 0.938 (0.611) | 0.913 (0.695) |
| In-degree | **1.102 (0.006)** | 1.091 (0.326) | 1.127 (0.114) | 1.069 (0.679) |
| Perceived Friend Use | **1.470 (0.000)** | **0.729 (0.022)** | **0.716 (0.002)** | 0.779 (0.319) |
| Network Exposure Users | **5.684 (0.000)** | **0.104 (0.011)** | **0.232 (0.031)** | 0.800 (0.878) |
| Network Exposure Dis-adopters | 1.814 (0.253) | 0.374 (0.271) | 0.401 (0.238) | 0.043 (0.323) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 1711 | 161 | 182 | 161 |
| N Events | 193 | 89 | 129 | 15 |

## 6.3 Q = 6

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 0.822 (0.213) | 1.891 (0.121) | 1.409 (0.377) | 0.452 (0.291) |
| Female | 1.295 (0.105) | 0.650 (0.210) | 0.859 (0.651) | 2.783 (0.158) |
| Sexual Minority | **1.379 (0.041)** | 0.762 (0.386) | 1.057 (0.853) | 2.193 (0.150) |
| Parent Ed. | 0.967 (0.541) | 0.998 (0.988) | 0.959 (0.753) | 0.789 (0.293) |
| Asian | **0.642 (0.016)** | 1.017 (0.966) | 1.018 (0.962) | 0.413 (0.193) |
| Hispanic/Latine | 1.148 (0.438) | 0.843 (0.646) | 0.850 (0.619) | 0.408 (0.157) |
| MDD (Major Depressive S.) | 1.119 (0.483) | 0.712 (0.273) | **0.510 (0.016)** | 0.351 (0.060) |
| GAD (Generalized Anxiety Dis.) | 0.889 (0.409) | 1.263 (0.415) | 1.162 (0.571) | 1.670 (0.337) |
| Out-degree | 0.938 (0.166) | 1.166 (0.183) | 1.004 (0.969) | 0.997 (0.988) |
| In-degree | **1.089 (0.008)** | 1.052 (0.500) | 1.091 (0.173) | 1.114 (0.409) |
| Perceived Friend Use | **1.502 (0.000)** | **0.720 (0.003)** | **0.719 (0.000)** | 0.798 (0.250) |
| Network Exposure Users | **5.835 (0.000)** | 0.237 (0.062) | 0.316 (0.068) | 0.720 (0.794) |
| Network Exposure Dis-adopters | 1.842 (0.184) | 0.336 (0.187) | 0.445 (0.250) | 0.160 (0.384) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 2077 | 206 | 233 | 206 |
| N Events | 232 | 108 | 159 | 21 |

## 6.4 Q = 5

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 0.831 (0.200) | 1.923 (0.080) | 1.428 (0.291) | 0.557 (0.300) |
| Female | **1.430 (0.017)** | 0.598 (0.115) | 0.781 (0.436) | 2.286 (0.171) |
| Sexual Minority | **1.432 (0.014)** | 0.865 (0.602) | 1.041 (0.879) | 1.580 (0.305) |
| Parent Ed. | 0.945 (0.274) | 0.985 (0.900) | 0.946 (0.648) | 0.830 (0.312) |
| Asian | **0.696 (0.037)** | 1.063 (0.871) | 1.003 (0.994) | 0.406 (0.130) |
| Hispanic/Latine | 1.184 (0.316) | 0.869 (0.676) | 0.878 (0.657) | 0.514 (0.187) |
| MDD (Major Depressive S.) | 1.136 (0.381) | 0.775 (0.371) | **0.531 (0.013)** | **0.379 (0.043)** |
| GAD (Generalized Anxiety Dis.) | 0.839 (0.187) | 1.207 (0.468) | 1.229 (0.391) | 1.525 (0.344) |
| Out-degree | 0.942 (0.173) | 1.133 (0.240) | 1.044 (0.646) | 1.028 (0.858) |
| In-degree | **1.082 (0.008)** | 1.047 (0.505) | 1.061 (0.311) | 1.051 (0.655) |
| Perceived Friend Use | **1.490 (0.000)** | **0.765 (0.007)** | **0.748 (0.000)** | 0.791 (0.124) |

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Network Exposure Users | **6.019 (0.000)** | 0.348 (0.123) | 0.521 (0.250) | 1.882 (0.470) |
| Network Exposure Dis-adopters | 2.039 (0.096) | 0.240 (0.071) | 0.336 (0.102) | 0.446 (0.614) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 2404 | 237 | 271 | 237 |
| N Events | 267 | 124 | 185 | 29 |

# 7. No $E_D$

§5 and §6 found $E_D$ to be noisy and never significant on A/B/C, while suggesting collinearity with $E_{\text{users}}$ when $E_D$ is the alt definition. §7 refits §5's four outcomes after **removing $E_D$ entirely** from the predictor list. The remaining 12 predictors are unchanged. Effective $N$ is identical to §5 (the rows lost to complete-cases are the same — $E_{\text{users}}$ and $E_D$ have correlated NAs). OR (p-value). Bold = $p < 0.05$.

## 7.1 Q = 8

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 1.139 (0.582) | 1.759 (0.426) | 0.902 (0.873) | 2.630 (0.537) |
| Female | 1.654 (0.058) | **0.337 (0.044)** | 0.414 (0.122) | 0.900 (0.951) |
| Sexual Minority | 1.539 (0.092) | 1.278 (0.687) | 1.704 (0.342) | **21.740 (0.041)** |
| Parent Ed. | 1.000 (0.998) | 1.164 (0.468) | 0.925 (0.769) | 0.305 (0.154) |
| Asian | 0.569 (0.050) | 1.198 (0.793) | 0.668 (0.491) | 0.283 (0.324) |
| Hispanic/Latine | 1.034 (0.907) | 0.741 (0.695) | 0.583 (0.405) | 0.010 (0.063) |
| MDD (Major Depressive S.) | 1.225 (0.420) | 0.502 (0.139) | **0.362 (0.020)** | **0.027 (0.017)** |
| GAD (Generalized Anxiety Dis.) | 0.876 (0.559) | 2.498 (0.076) | 1.858 (0.208) | 9.083 (0.126) |
| Out-degree | **0.828 (0.014)** | 1.359 (0.106) | 1.032 (0.874) | 0.648 (0.431) |
| In-degree | **1.106 (0.038)** | 1.124 (0.405) | 1.020 (0.876) | 0.541 (0.242) |
| Perceived Friend Use | **1.470 (0.000)** | 0.906 (0.661) | 0.925 (0.650) | 1.824 (0.329) |
| Network Exposure Users | **3.658 (0.036)** | **0.021 (0.019)** | **0.066 (0.008)** | 0.043 (0.329) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 925 | 83 | 91 | 83 |
| N Events | 89 | 52 | 70 | 7 |

## 7.2 Q = 7

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 0.863 (0.374) | 1.605 (0.296) | 1.180 (0.702) | 0.531 (0.436) |
| Female | 1.352 (0.098) | 0.781 (0.526) | 1.076 (0.845) | 3.113 (0.164) |
| Sexual Minority | **1.461 (0.028)** | 0.700 (0.330) | 0.857 (0.665) | 1.163 (0.819) |
| Parent Ed. | 0.973 (0.641) | 0.994 (0.963) | 0.862 (0.344) | 0.582 (0.065) |
| Asian | **0.618 (0.014)** | 0.947 (0.904) | 1.033 (0.934) | 0.402 (0.238) |
| Hispanic/Latine | 1.074 (0.714) | 0.808 (0.627) | 0.691 (0.316) | 0.273 (0.103) |
| MDD (Major Depressive S.) | 1.167 (0.378) | 0.753 (0.396) | **0.454 (0.007)** | **0.217 (0.025)** |
| GAD (Generalized Anxiety Dis.) | 0.865 (0.351) | 1.386 (0.315) | 1.187 (0.562) | 1.305 (0.675) |
| Out-degree | 0.913 (0.074) | 1.173 (0.227) | 0.940 (0.618) | 0.912 (0.693) |
| In-degree | **1.101 (0.007)** | 1.097 (0.291) | 1.128 (0.110) | 1.075 (0.655) |

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Perceived Friend Use | **1.478 (0.000)** | **0.706 (0.011)** | **0.696 (0.001)** | 0.744 (0.234) |
| Network Exposure Users | **5.307 (0.000)** | **0.135 (0.018)** | 0.297 (0.058) | 1.178 (0.907) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 1711 | 161 | 182 | 161 |
| N Events | 193 | 89 | 129 | 15 |

## 7.3 Q = 6

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 0.829 (0.231) | 1.793 (0.150) | 1.387 (0.395) | 0.428 (0.257) |
| Female | 1.307 (0.091) | 0.611 (0.148) | 0.823 (0.557) | 2.680 (0.175) |
| Sexual Minority | **1.387 (0.037)** | 0.773 (0.414) | 1.082 (0.793) | 2.146 (0.162) |
| Parent Ed. | 0.967 (0.543) | 0.981 (0.870) | 0.951 (0.699) | 0.790 (0.296) |
| Asian | **0.617 (0.007)** | 1.072 (0.859) | 1.084 (0.826) | 0.469 (0.253) |
| Hispanic/Latine | 1.128 (0.493) | 0.825 (0.600) | 0.868 (0.659) | 0.435 (0.181) |
| MDD (Major Depressive S.) | 1.113 (0.502) | 0.730 (0.309) | **0.517 (0.018)** | 0.360 (0.067) |
| GAD (Generalized Anxiety Dis.) | 0.890 (0.417) | 1.221 (0.484) | 1.143 (0.615) | 1.600 (0.380) |
| Out-degree | 0.935 (0.152) | 1.163 (0.193) | 1.009 (0.936) | 0.994 (0.976) |
| In-degree | **1.089 (0.008)** | 1.052 (0.494) | 1.091 (0.177) | 1.107 (0.437) |
| Perceived Friend Use | **1.509 (0.000)** | **0.699 (0.001)** | **0.704 (0.000)** | 0.763 (0.154) |
| Network Exposure Users | **5.425 (0.000)** | 0.320 (0.109) | 0.401 (0.116) | 1.034 (0.977) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 2077 | 206 | 233 | 206 |
| N Events | 232 | 108 | 159 | 21 |

## 7.4 Q = 5

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 0.840 (0.224) | 1.768 (0.120) | 1.377 (0.339) | 0.541 (0.274) |
| Female | **1.446 (0.013)** | 0.551 (0.065) | 0.731 (0.316) | 2.222 (0.185) |
| Sexual Minority | **1.440 (0.012)** | 0.867 (0.610) | 1.067 (0.807) | 1.563 (0.315) |
| Parent Ed. | 0.945 (0.275) | 0.970 (0.794) | 0.936 (0.579) | 0.831 (0.317) |
| Asian | **0.665 (0.017)** | 1.126 (0.748) | 1.085 (0.809) | 0.424 (0.145) |
| Hispanic/Latine | 1.159 (0.374) | 0.844 (0.604) | 0.903 (0.724) | 0.520 (0.192) |
| MDD (Major Depressive S.) | 1.131 (0.398) | 0.797 (0.421) | **0.536 (0.015)** | **0.384 (0.045)** |
| GAD (Generalized Anxiety Dis.) | 0.841 (0.192) | 1.176 (0.526) | 1.214 (0.427) | 1.504 (0.361) |
| Out-degree | 0.939 (0.156) | 1.134 (0.238) | 1.054 (0.572) | 1.030 (0.851) |
| In-degree | **1.081 (0.008)** | 1.046 (0.514) | 1.059 (0.324) | 1.047 (0.678) |
| Perceived Friend Use | **1.496 (0.000)** | **0.739 (0.002)** | **0.730 (0.000)** | 0.777 (0.090) |
| Network Exposure Users | **5.543 (0.000)** | 0.496 (0.260) | 0.694 (0.488) | 2.188 (0.342) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 2404 | 237 | 271 | 237 |
| N Events | 267 | 124 | 185 | 29 |

**Reading.** Dropping $E_D$ leaves the §5 picture essentially intact: PFU and $E_{\text{users}}$ are still the two robust significant predictors of disadoption at $Q \leq 7$, both with OR $< 1$ on A and B (consistent with peer-use environment retaining vapers, not pushing

them out). The marginal change relative to §5 is small — the $E_{\text{users}}$ ORs *don't shift much* (e.g., A at Q=7: 0.099→0.095; A at Q=8: 0.021→0.026), confirming that $E_D$ peer-flipped (§5) was not absorbing variance from $E_{\text{users}}$. The intuitive lesson is **the peer-use environment effect on disadoption is carried entirely by $E_{\text{users}}$** ("how many of your friends currently use") rather than by any disadoption-specific peer signal: knowing the share of friends *currently using* is sufficient — the share who *recently quit* adds nothing once $E_{\text{users}}$ is in the model. By contrast, in §6 (alt $E_D$), the alt-$E_D$ definition is mechanically a function of $E_{\text{users}}$ ($E^{\max} - E_{\text{current}}$), so it shifts variance off $E_{\text{users}}$ and onto itself — that's why $E_{\text{users}}$ looks slightly cleaner in §6 than in §7. The cleanest specification, and the one we recommend as the headline interpretation, is therefore §7: **drop $E_D$, keep $E_{\text{users}}$, and read the disadoption-side coefficient as the peer-use channel.**

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# 8. Model C window sensitivity

Three columns per Q: $C$ at $W = 1$ (immediate return), $W \leq 2$, $W \leq 3$.

**Event-count summary** (number of cyclic 1→0 events for the same Q-eligible sample):

| Q | C, W=1 | C, W 2 | C, W 3 | Δ(W 2 − W=1) | Δ(W 3 − W=2) |
|---|---|---|---|---|---|
| 8 | 17 | 27 | 29 | +10 | +2 |
| 7 | 43 | 63 | 68 | +20 | +5 |
| 6 | 66 | 92 | 97 | +26 | +5 |
| 5 | 81 | 114 | 121 | +33 | +7 |

The big increment is from $W = 1 \to W \leq 2$ (~30–40% more events); $W \leq 3$ adds little beyond $W \leq 2$.

## 8.1 Q = 8

| Variable | C, W=1 | C, W 2 | C, W 3 |
|---|---|---|---|
| Cohort (2025 vs 2024) | 2.248 (0.613) | 6.829 (0.969) | 6.829 (0.969) |
| Female | 0.855 (0.926) | 7.415 (0.963) | 7.415 (0.963) |
| Sexual Minority | **21.702 (0.040)** | 40.169 (0.900) | 40.169 (0.900) |
| Parent Ed. | 0.327 (0.175) | 0.314 (0.955) | 0.314 (0.955) |
| Asian | 0.264 (0.299) | 0.056 (0.934) | 0.056 (0.934) |
| Hispanic/Latine | 0.009 (0.059) | 0.036 (0.916) | 0.036 (0.916) |
| MDD (Major Depressive S.) | **0.028 (0.019)** | 0.006 (0.887) | 0.006 (0.887) |
| GAD (Generalized Anxiety Dis.) | 10.328 (0.129) | 1.976 (0.982) | 1.976 (0.982) |
| Out-degree | 0.683 (0.485) | 0.290 (0.942) | 0.290 (0.942) |
| In-degree | 0.556 (0.268) | 1.214 (0.982) | 1.214 (0.982) |
| Perceived Friend Use | 1.843 (0.329) | 0.795 (0.984) | 0.795 (0.984) |
| Network Exposure Users | 0.080 (0.447) | 0.950 (1.000) | 0.950 (1.000) |
| Network Exposure Dis-adopters | 0.059 (0.691) | 1681.502 (0.934) | 1681.502 (0.934) |
| Rho (ICC) | 0.000 | 0.994 | 0.994 |
| N Students | 83 | 83 | 83 |
| N Events | 7 | 11 | 11 |

## 8.2 Q = 7

| Variable | C, W=1 | C, W 2 | C, W 3 |
|---|---|---|---|
| Cohort (2025 vs 2024) | 0.535 (0.444) | 0.176 (0.824) | 0.280 (0.871) |
| Female | 3.125 (0.163) | 1.964 (0.932) | 3.902 (0.865) |
| Sexual Minority | 1.165 (0.817) | 4.198 (0.803) | 6.801 (0.744) |
| Parent Ed. | 0.584 (0.068) | 0.379 (0.749) | 0.336 (0.704) |
| Asian | 0.401 (0.237) | 0.453 (0.933) | 1.237 (0.981) |
| Hispanic/Latine | 0.274 (0.104) | 0.663 (0.964) | 0.228 (0.877) |
| MDD (Major Depressive S.) | **0.217 (0.025)** | 0.108 (0.685) | 0.029 (0.528) |

| Variable | C, W=1 | C, W 2 | C, W 3 |
|---|---|---|---|
| GAD (Generalized Anxiety Dis.) | 1.305 (0.676) | 0.363 (0.830) | 0.357 (0.838) |
| Out-degree | 0.915 (0.705) | 0.675 (0.821) | 0.465 (0.682) |
| In-degree | 1.075 (0.654) | 1.170 (0.931) | 1.279 (0.889) |
| Perceived Friend Use | 0.748 (0.245) | 0.354 (0.597) | 0.414 (0.646) |
| Network Exposure Users | 1.184 (0.903) | 17.109 (0.845) | 2.213 (0.955) |
| Network Exposure Dis-adopters | 0.704 (0.896) | 11.801 (0.898) | 62.406 (0.832) |
| Rho (ICC) | 0.000 | 0.982 | 0.983 |
| N Students | 161 | 161 | 161 |
| N Events | 15 | 20 | 21 |

## 8.3 Q = 6

| Variable | C, W=1 | C, W 2 | C, W 3 |
|---|---|---|---|
| Cohort (2025 vs 2024) | 0.431 (0.262) | 1.021 (0.972) | 0.920 (0.887) |
| Female | 2.669 (0.177) | 1.640 (0.417) | 1.827 (0.318) |
| Sexual Minority | 2.152 (0.160) | 2.435 (0.070) | 2.316 (0.082) |
| Parent Ed. | 0.792 (0.301) | 0.820 (0.326) | 0.875 (0.484) |
| Asian | 0.468 (0.252) | 0.329 (0.081) | 0.454 (0.189) |
| Hispanic/Latine | 0.436 (0.183) | 0.565 (0.296) | 0.549 (0.270) |
| MDD (Major Depressive S.) | 0.361 (0.068) | 0.533 (0.196) | 0.471 (0.121) |
| GAD (Generalized Anxiety Dis.) | 1.597 (0.381) | 1.039 (0.936) | 1.041 (0.934) |
| Out-degree | 0.996 (0.985) | 0.931 (0.674) | 0.923 (0.633) |
| In-degree | 1.106 (0.439) | 1.160 (0.217) | 1.128 (0.302) |
| Perceived Friend Use | 0.764 (0.158) | 0.818 (0.230) | 0.847 (0.304) |
| Network Exposure Users | 1.038 (0.975) | 0.606 (0.646) | 0.951 (0.961) |
| Network Exposure Dis-adopters | 0.702 (0.865) | 2.916 (0.425) | 3.028 (0.409) |
| Rho (ICC) | 0.000 | 0.000 | 0.000 |
| N Students | 206 | 206 | 206 |
| N Events | 21 | 26 | 27 |

## 8.4 Q = 5

| Variable | C, W=1 | C, W 2 | C, W 3 |
|---|---|---|---|
| Cohort (2025 vs 2024) | 0.557 (0.301) | 0.858 (0.754) | 0.783 (0.616) |
| Female | 2.198 (0.193) | 1.568 (0.402) | 1.656 (0.344) |
| Sexual Minority | 1.596 (0.295) | 1.528 (0.301) | 1.493 (0.324) |
| Parent Ed. | 0.838 (0.339) | 0.843 (0.308) | 0.882 (0.431) |
| Asian | 0.424 (0.145) | 0.337 (0.056) | 0.441 (0.131) |
| Hispanic/Latine | 0.533 (0.211) | 0.608 (0.275) | 0.603 (0.265) |
| MDD (Major Depressive S.) | **0.382 (0.045)** | 0.541 (0.147) | 0.495 (0.097) |
| GAD (Generalized Anxiety Dis.) | 1.501 (0.362) | 1.230 (0.610) | 1.245 (0.589) |
| Out-degree | 1.038 (0.813) | 1.056 (0.698) | 1.053 (0.711) |
| In-degree | 1.047 (0.683) | 1.037 (0.726) | 1.018 (0.860) |
| Perceived Friend Use | 0.787 (0.110) | 0.821 (0.150) | 0.841 (0.198) |
| Network Exposure Users | 2.122 (0.359) | 1.444 (0.639) | 1.879 (0.405) |
| Network Exposure Dis-adopters | 0.190 (0.430) | 1.185 (0.899) | 1.200 (0.891) |
| Rho (ICC) | 0.000 | 0.000 | 0.000 |
| N Students | 237 | 237 | 237 |
| N Events | 29 | 35 | 36 |

# 9. Sensitivity (a) — indeterminates counted as Stable (A)

In §5/§6/§8/§10, $1 \to 0$ events with NA-only future are dropped from A (we cannot verify "no return"). §9 instead **counts them as A events** (assumes the student would not have returned). Adopters / B / C are unchanged from §5; only A's panel grows. Two tables, one per $E_D$ definition. Each table has paired columns per Q level: **(orig)** = §5/§6 main A; **(a)** = indeterminate $1 \to 0$ counted as A.

**Event-count summary** (A column only; rest unchanged from §5):

| Q | A events (orig) | A events (a) | Δ |
|---|---|---|---|
| 8 | 52 | 66 | +14 (+27%) |
| 7 | 89 | 122 | +33 (+37%) |
| 6 | 108 | 150 | +42 (+39%) |
| 5 | 124 | 169 | +45 (+36%) |

## 9.1 Table — $E_D$ = peer-flipped $1 \to 0$

| Variable | Q=8 (orig) | Q=8 (a) | Q=7 (orig) | Q=7 (a) | Q=6 (orig) | Q=6 (a) | Q=5 (orig) | Q=5 (a) |
|---|---|---|---|---|---|---|---|---|
| Cohort (2025 vs 2024) | 1.771 (0.424) | 1.009 (0.987) | 1.705 (0.247) | 1.379 (0.426) | 1.851 (0.133) | 1.553 (0.228) | 1.786 (0.116) | 1.599 (0.146) |
| Female | **0.336 (0.044)** | 0.519 (0.216) | 0.760 (0.482) | 0.952 (0.890) | 0.596 (0.130) | 0.850 (0.604) | 0.547 (0.062) | 0.770 (0.377) |
| Sexual Minority | 1.279 (0.687) | 0.852 (0.757) | 0.734 (0.406) | 0.756 (0.403) | 0.790 (0.456) | 0.772 (0.364) | 0.877 (0.639) | 0.847 (0.510) |
| Parent Ed. | 1.158 (0.496) | 1.111 (0.614) | 1.014 (0.922) | 0.956 (0.737) | 0.987 (0.911) | 0.988 (0.915) | 0.972 (0.811) | 0.976 (0.829) |
| Asian | 1.188 (0.800) | 1.009 (0.987) | 0.979 (0.963) | 1.202 (0.623) | 1.089 (0.827) | 1.180 (0.621) | 1.133 (0.735) | 1.161 (0.636) |
| Hispanic/Latino | 0.724 (0.668) | 1.076 (0.896) | 0.882 (0.781) | 0.991 (0.980) | 0.863 (0.690) | 0.963 (0.902) | 0.858 (0.642) | 0.961 (0.886) |
| MDD (Major Depressive S.) | 0.502 (0.140) | 0.649 (0.262) | 0.741 (0.380) | 0.703 (0.256) | 0.728 (0.306) | 0.666 (0.152) | 0.795 (0.415) | 0.708 (0.182) |
| GAD (Generalized Anxiety Dis.) | 2.470 (0.079) | 1.340 (0.494) | 1.451 (0.264) | 1.052 (0.866) | 1.241 (0.453) | 1.008 (0.976) | 1.181 (0.517) | 1.011 (0.964) |
| Out-degree | 1.356 (0.109) | 1.268 (0.166) | 1.184 (0.204) | 1.055 (0.656) | 1.168 (0.177) | 1.053 (0.609) | 1.135 (0.236) | 1.051 (0.596) |
| In-degree | 1.124 (0.406) | 1.028 (0.812) | 1.103 (0.265) | 1.094 (0.227) | 1.052 (0.499) | 1.078 (0.237) | 1.045 (0.520) | 1.080 (0.194) |
| Perceived Friend Use | 0.904 (0.657) | 0.897 (0.504) | **0.710 (0.013)** | **0.708 (0.001)** | **0.700 (0.001)** | **0.715 (0.000)** | **0.742 (0.002)** | **0.751 (0.000)** |
| Network Exposure Users | **0.021 (0.018)** | **0.080 (0.044)** | **0.135 (0.016)** | 0.312 (0.075) | 0.324 (0.110) | 0.456 (0.176) | 0.491 (0.253) | 0.570 (0.295) |
| Network Exposure Dis-adopters | 1.211 (0.892) | 1.808 (0.616) | 0.164 (0.169) | 0.929 (0.921) | 0.374 (0.339) | 1.332 (0.677) | 0.645 (0.651) | 1.589 (0.441) |
| N Students | 83 | 96 | 161 | 191 | 206 | 245 | 237 | 283 |
| N Events | 52 | 66 | 89 | 122 | 108 | 150 | 124 | 169 |

## 9.2 Table — $E_D = E^{\max} - E_{\textbf{current}}$ (alt)

| Variable | Q=8 (orig) | Q=8 (a) | Q=7 (orig) | Q=7 (a) | Q=6 (orig) | Q=6 (a) | Q=5 (orig) | Q=5 (a) |
|---|---|---|---|---|---|---|---|---|
| Cohort (2025 vs 2024) | 1.771 (0.418) | 0.982 (0.975) | 1.682 (0.255) | 1.383 (0.422) | 1.891 (0.121) | 1.560 (0.224) | 1.923 (0.080) | 1.644 (0.130) |
| Female | **0.344 (0.045)** | 0.512 (0.205) | 0.834 (0.649) | 0.953 (0.896) | 0.650 (0.210) | 0.852 (0.613) | 0.598 (0.115) | 0.793 (0.439) |
| Sexual Minority | 1.264 (0.699) | 0.854 (0.761) | 0.682 (0.297) | 0.755 (0.403) | 0.762 (0.386) | 0.773 (0.368) | 0.865 (0.602) | 0.848 (0.515) |
| Parent Ed. | 1.179 (0.426) | 1.121 (0.575) | 1.025 (0.856) | 0.956 (0.735) | 0.998 (0.988) | 0.992 (0.941) | 0.985 (0.900) | 0.987 (0.908) |
| Asian | 1.189 (0.803) | 1.038 (0.945) | 0.913 (0.844) | 1.197 (0.635) | 1.017 (0.966) | 1.180 (0.630) | 1.063 (0.871) | 1.124 (0.718) |
| Hispanic/Latino | 0.757 (0.716) | 1.121 (0.840) | 0.826 (0.668) | 0.989 (0.975) | 0.843 (0.646) | 0.972 (0.926) | 0.869 (0.676) | 0.961 (0.890) |
| MDD (Major Depressive S.) | 0.498 (0.132) | 0.654 (0.267) | 0.736 (0.365) | 0.702 (0.257) | 0.712 (0.273) | 0.666 (0.153) | 0.775 (0.371) | 0.706 (0.184) |
| GAD (Generalized Anxiety Dis.) | 2.535 (0.080) | 1.359 (0.475) | 1.427 (0.279) | 1.053 (0.863) | 1.263 (0.415) | 1.008 (0.976) | 1.207 (0.468) | 1.008 (0.973) |
| Outdegree | 1.367 (0.109) | 1.264 (0.176) | 1.183 (0.209) | 1.056 (0.647) | 1.166 (0.183) | 1.051 (0.619) | 1.133 (0.240) | 1.042 (0.658) |
| Indegree | 1.120 (0.430) | 1.029 (0.807) | 1.091 (0.326) | 1.094 (0.228) | 1.052 (0.500) | 1.077 (0.241) | 1.047 (0.505) | 1.080 (0.198) |
| Perceived Friend Use | 0.910 (0.674) | 0.900 (0.522) | **0.729 (0.022)** | **0.708 (0.001)** | **0.720 (0.003)** | **0.719 (0.000)** | **0.765 (0.007)** | **0.767 (0.001)** |
| Network Exposure Users | **0.019 (0.020)** | 0.088 (0.060) | **0.104 (0.011)** | 0.310 (0.092) | 0.237 (0.062) | 0.439 (0.196) | 0.348 (0.123) | 0.481 (0.206) |
| Network Exposure Disadopters | 0.723 (0.818) | 1.153 (0.900) | 0.374 (0.271) | 0.971 (0.969) | 0.336 (0.187) | 0.867 (0.829) | 0.240 (0.071) | 0.572 (0.363) |
| N Students | 83 | 96 | 161 | 191 | 206 | 245 | 237 | 283 |
| N Events | 52 | 66 | 89 | 122 | 108 | 150 | 124 | 169 |

**Reading**. Switching from (orig) to (a) within either $E_D$ definition adds 27–39% more A events but does not change *which* predictors are significant: PFU and $E_{\text{users}}$ remain the only consistent disadoption predictors at Q=7 and below; both predictors push toward less disadoption (OR $< 1$), as expected if peer-use environments stabilise smoking once initiated. The two tables differ visibly only on $E_D$ itself: peer-flipped $E_D$ is noisy and never significant; the alt definition has tighter point estimates but is equally non-significant at conventional levels. See §11 for our reading.

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# 10. Sensitivity (b) — observed-wave jumps

§10 walks through observed waves regardless of calendar gaps (instead of consecutive-calendar pairs).

**Event-count summary** vs §5 main:

| Q | adopt (§5) | adopt (§10) | A (§5) | A (§10) | B (§5) | B (§10) | C (§5) | C (§10) |
|---|---|---|---|---|---|---|---|---|
| 8 | 162 | 162 | 96 | 96 | 142 | 142 | 17 | 17 |
| 7 | 346 | 346 | 189 | 189 | 293 | 293 | 43 | 43 |
| 6 | 449 | 449 | 244 | 244 | 384 | 384 | 66 | 66 |
| 5 | 551 | 551 | 288 | 288 | 467 | 465 | 81 | 81 |

The W1-W10 panel has very few NA gaps within consecutive observations; observed-wave jumps and calendar-pair logic produce nearly identical event sets (only B at Q=5 differs by 2 events). Tables §10.1–§10.4 are therefore visually indistinguishable from §5.1–§5.4 and we reproduce only Q = 5 here for reference; the full set is in `outputs/tables/v4b_table_9_Q{8,7,6,5}.csv`.

## 10.1 Q = 5 (representative)

| Variable | Adopters | A | B | C |
|---|---|---|---|---|
| Cohort (2025 vs 2024) | 0.836 (0.213) | 1.786 (0.116) | 1.379 (0.337) | 0.557 (0.301) |
| Female | **1.431 (0.016)** | 0.547 (0.062) | 0.732 (0.317) | 2.198 (0.193) |
| Sexual Minority | **1.435 (0.013)** | 0.877 (0.639) | 1.060 (0.826) | 1.596 (0.295) |
| Parent Ed. | 0.948 (0.301) | 0.972 (0.811) | 0.935 (0.575) | 0.838 (0.339) |
| Asian | **0.678 (0.022)** | 1.133 (0.735) | 1.086 (0.809) | 0.424 (0.145) |
| Hispanic/Latine | 1.162 (0.362) | 0.858 (0.642) | 0.905 (0.732) | 0.533 (0.211) |
| MDD (Major Depressive S.) | 1.137 (0.375) | 0.795 (0.415) | **0.536 (0.015)** | **0.382 (0.045)** |
| GAD (Generalized Anxiety Dis.) | 0.840 (0.190) | 1.181 (0.517) | 1.217 (0.422) | 1.501 (0.362) |
| Out-degree | 0.941 (0.164) | 1.135 (0.236) | 1.057 (0.562) | 1.038 (0.813) |
| In-degree | **1.081 (0.009)** | 1.045 (0.520) | 1.060 (0.322) | 1.047 (0.683) |
| Perceived Friend Use | **1.490 (0.000)** | **0.742 (0.002)** | **0.729 (0.000)** | 0.787 (0.110) |
| Network Exposure Users | **5.699 (0.000)** | 0.491 (0.253) | 0.701 (0.502) | 2.122 (0.359) |
| Network Exposure Dis-adopters | 2.926 (0.058) | 0.645 (0.651) | 1.184 (0.803) | 0.190 (0.430) |
| Rho (ICC) | — | — | — | 0.000 |
| N Students | 2404 | 237 | 271 | 237 |
| N Events | 267 | 124 | 185 | 29 |

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# 11. Descriptive associations and sensitivities

The §5/§9 regressions show **Perceived Friend Use (PFU)** as the cleanest two-sided lever for adoption, but the disadoption coefficients (although consistently negative on A and B) sometimes look modest after adjusting for 12 other predictors. To sanity-check the raw signal we look at unadjusted associations on the W1-W10 panel.

## 11.1 Out-degree distribution by ego status

Out-degree = number of friends an ego nominated whose nomination survives the panel-hygiene rules (alter in panel and responding at the same wave). The maximum out-degree is 7 by questionnaire design.

| Out-degree | n (all ego-waves) | n (current user, ecig=1) | n (current non-user, ecig=0) |
|---|---|---|---|
| 1 | 2,956 | 235 | 2,716 |
| 2 | 3,815 | 267 | 3,541 |
| 3 | 4,195 | 264 | 3,922 |
| 4 | 3,840 | 228 | 3,602 |
| 5 | 2,708 | 153 | 2,550 |
| 6 | 1,533 | 60 | 1,470 |
| 7 | 481 | 24 | 456 |
| **Total** | **19,528** | **1,231** | **18,257** |

Mean out-degree: 3.31 (all) / 3.06 (users) / 3.33 (non-users). Median is 3 in all three subsets.

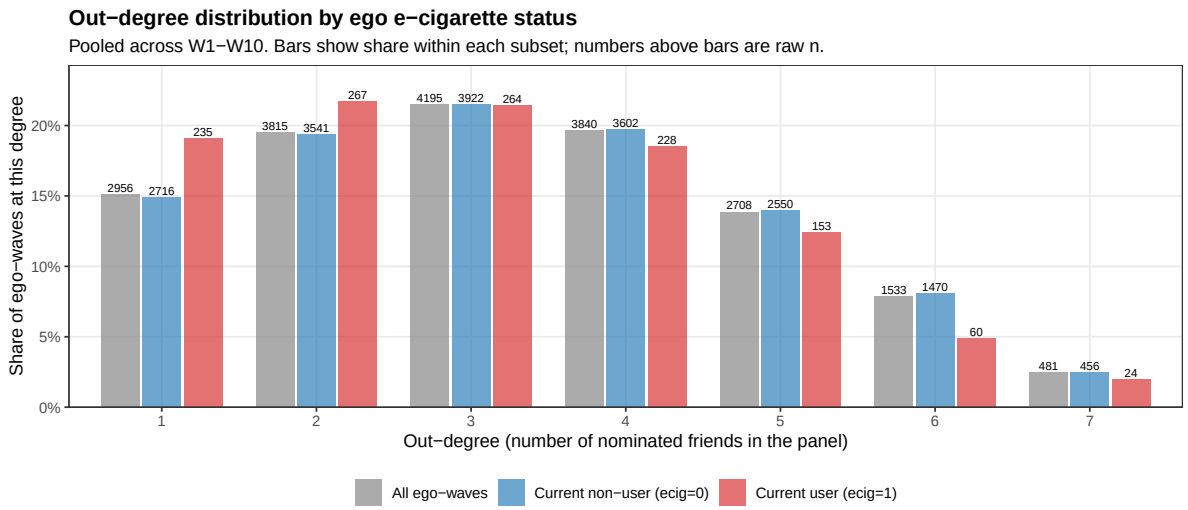



Figure 5: Out-degree distribution by ego e-cigarette status. Bars dodge per subset; share within subset on the y-axis, raw n on top of each bar.

**Reading.** Users are slightly *less* connected than non-users: the user distribution leans toward out-degree 1–3 (62.6% of users vs 56.9% of non-users), and away from out-degree 5–7 (19.2% of users vs 24.5% of non-users). The mean out-degree gap is 0.27 friends. This is a small but substantively interesting fact — a higher-density friendship network is not what predicts ecig use here; rather, *who* the friends are (PFU, $E_{\text{users}}$) matters. The distribution shape is very stable across the three subsets, which is reassuring for the network exposure measures: $E_{\text{users}}$ is computed on a denominator that varies little between users and non-users.

## 11.2 Event rate by PFU at $w-1$

For each person-wave row eligible for the corresponding risk-set (no Q-restriction; all W1-W10 panel rows with valid lag and outcome). **Disadoption here is *any*** $1 \to 0$ transition (Model B-style: $\text{ecig}_{w-1} = 1$ and $\text{ecig}_{w} = 0$, regardless of return). It is **not** the stable-A definition — there is no requirement that the student stay at 0 in later observed waves.

| PFU at $w-1$ | At risk (adopt) | N adopted | Adoption rate | At risk (disad) | N disadopted | Disadoption rate |
|---|---|---|---|---|---|---|
| 0 (None) | 17,492 | 376 | 2.15 % | 388 | 210 | **54.12 %** |
| 1 | 2,067 | 106 | 5.13 % | 224 | 95 | 42.41 % |
| 2 | 1,246 | 111 | 8.91 % | 247 | 77 | 31.17 % |
| 3 | 652 | 62 | 9.51 % | 223 | 61 | 27.35 % |
| 4 | 214 | 20 | 9.35 % | 127 | 27 | 21.26 % |
| 5 | 269 | 31 | **11.52 %** | 193 | 40 | 20.73 % |

PFU at $w-1$ moves adoption from 2% $\to$ 12% (5.4×) and **moves disadoption from 54% $\to$ 21%** (2.6× lower). Point-biserial correlations with the binary outcomes:

| Outcome (binary at-risk row) | $r$ | n |
|---|---|---|
| Adoption (any 0→1) | **+0.128** | 21,940 |
| Any 1→0 transition | **−0.256** | 1,402 |

The disadoption signal is large in raw form. Its visibility in §5/§6/§9 depends on which sub-event we model (A / B / C), the small sub-samples, and the competition with `Network Exposure Users`.

**Same exercise but with the network-derived measure $E_{\text{users}}$ at $w-1$** (peer share who currently use ecig, computed from the friendship-nomination network — *not* the self-report PFU). $E_{\text{users}}$ is binned into 6 categories, parallel to the 0–5 PFU scale. Same denominator definitions as above; "Disadoption rate" is again *any* $1 \to 0$.

| $E_{\text{users}}$ at $w-1$ | At risk (adopt) | N adopted | Adoption rate | At risk (disad) | N disadopted | Disadoption rate |
|---|---|---|---|---|---|---|
| 0 (None) | 12,223 | 384 | 3.14 % | 537 | 280 | 52.14 % |
| (0, 0.2] | 683 | 35 | 5.12 % | 72 | 42 | **58.33 %** |
| (0.2, 0.4] | 1,119 | 99 | 8.85 % | 160 | 72 | 45.00 % |
| (0.4, 0.6] | 406 | 38 | 9.36 % | 101 | 33 | 32.67 % |
| (0.6, 0.8] | 74 | 9 | **12.16 %** | 30 | 13 | 43.33 % |
| (0.8, 1.0] | 136 | 11 | 8.09 % | 37 | 11 | 29.73 % |

| Outcome (binary at-risk row) | $r$ | n |
|---|---|---|
| Adoption (any 0→1) | **+0.090** | 14,641 |
| Any 1→0 transition | **−0.137** | 937 |

The qualitative pattern matches PFU — disadoption falls from 52% at $E_{\text{users}} = 0$ to 30% at $E_{\text{users}} > 0.8$ (1.7× lower) — but the gradient is gentler and noisier than the self-reported PFU gradient (54.1% → 20.7%, 2.6×). The $(0, 0.2]$ bin is anomalous (58.3%), but that's just 72 person-waves and overlaps with the 52% in the "none" bin within sampling noise. The point-biserial correlations are correspondingly weaker ($|r| = 0.137$ vs 0.256 for PFU). Two reasons: (i) $E_{\text{users}}$ has a much heavier mass at exactly 0 (12,223 vs 17,492 person-waves), and (ii) sparse out-degree means $E_{\text{users}}$ is a noisy estimator of "what fraction of friends use" — *self-reported* PFU averages over a wider, more salient personal network than the (small) friendship-nomination set captures.

**Same exercise but with the network-derived measure as a *count* of using friends**. We bin $k_{\text{users}} = E_{\text{users}} \times$ out_degree (rounded to integer) — i.e. how many friends the ego nominated who currently use ecig. We add explicit event-count columns so the rate is fully auditable per cell.

| # using friends $w-1$ | At risk (adopt) | N adopted | Adoption rate | At risk (disad) | N disadopted | Disadoption rate |
|---|---|---|---|---|---|---|
| 0 | 12,223 | 384 | 3.14 % | 537 | 280 | **52.14 %** |
| 1 | 2,024 | 154 | 7.61 % | 285 | 130 | 45.61 % |
| 2 | 342 | 31 | 9.06 % | 91 | 34 | 37.36 % |
| 3 | 45 | 6 | **13.33 %** | 17 | 5 | 29.41 % |
| 4 | 6 | 1 | 16.67 % | 5 | 2 | 40.00 % |
| 5+ | 1 | 0 | 0.00 % | 2 | 0 | 0.00 % |

| Outcome (binary at-risk row) | $r$ vs $k_{\text{users}}$ | n |
|---|---|---|
| Adoption (any 0→1) | **+0.092** | 14,641 |
| Any 1→0 transition | **−0.113** | 937 |

Reading the count version directly: a single using friend cuts disadoption from 52% to 46%; two using friends cuts it to 37%; three to 29%. The downward gradient is monotonic from $k = 0$ to $k = 3$ (where most of the

data sit) and then becomes erratic at $k \geq 4$ because there are very few egos with that many using friends per wave (51 person-waves total above $k = 3$). Same caveat as the proportion table: $E_{\text{users}}$ is a noisier proxy of peer-use environment than self-reported PFU, because the friendship-nomination network averages over only ~3 alters per ego.

## 11.3 PFU vs network exposures (Pearson)

PFU is correlated with the network-derived exposure measures, as expected:

| Pair (both at $w-1$) | $r$ | n |
|---|---|---|
| PFU vs $E_{\text{users}}$ | **+0.226** | 17,456 |
| PFU vs $E_D$ (peer-flipped 1→0) | +0.071 | 17,465 |
| PFU vs $E_D$ alt (peak − current) | +0.108 | 17,456 |
| PFU vs out-degree | −0.083 | 23,391 |
| PFU vs in-degree | −0.057 | 23,391 |

PFU and $E_{\text{users}}$ overlap ($\sim r = 0.23$) — both measure peer-user environment. They are not redundant: PFU is self-reported about close friends; $E_{\text{users}}$ is computed from the friendship-nomination network. The §5 regressions show both surviving as significant predictors of adoption (PFU OR ≈ 1.47–1.49, $E_{\text{users}}$ OR ≈ 5.3–5.9), suggesting they capture complementary aspects of peer-user salience.

## 11.4 Event rate by HS grade-semester

Grade-semester is derived from (cohort, wave) on the standard ADVANCE timeline. Eight semesters span fall 9th (gs=1) through spring 12th (gs=8):

- **Class of 2024** (schools 101–114): W1=fall 9th (gs=1), W2=spring 9th (gs=2), …, W8=spring 12th (gs=8). W9–W10 are post-HS and excluded.
- **Class of 2025** (schools 201–214): W3=gs=1, W4=gs=2, …, W10=gs=8.

The first observed wave for each cohort produces no at-risk rows because the lag is undefined, so gs=1 (fall 9th) has zero rows and is omitted; the table starts at gs=2 (spring 9th).

| Grade-semester | At risk (adopt) | N adopted | Adoption rate | At risk (disad) | N disadopted | Disadoption rate (any 1→0) |
|---|---|---|---|---|---|---|
| 2 (spring 9th) | 1,787 | 47 | 2.63 % | 45 | 21 | 46.67 % |
| 3 (fall 10th) | 2,865 | 126 | 4.40 % | 111 | 43 | 38.74 % |
| 4 (spring 10th) | 3,209 | 172 | **5.36 %** | 211 | 99 | 46.92 % |
| 5 (fall 11th) | 3,157 | 150 | 4.75 % | 273 | 138 | 50.55 % |
| 6 (spring 11th) | 2,977 | 166 | **5.58 %** | 262 | 129 | 49.24 % |
| 7 (fall 12th) | 2,715 | 100 | 3.68 % | 249 | 130 | **52.21 %** |
| 8 (spring 12th) | 2,620 | 81 | 3.09 % | 208 | 104 | 50.00 % |

| Outcome (binary at-risk row) | $r$ (Pearson) | n |
|---|---|---|
| Adoption (any 0→1) vs grade-semester | −0.006 | 19,330 |
| Any 1→0 transition vs grade-semester | +0.048 | 1,359 |

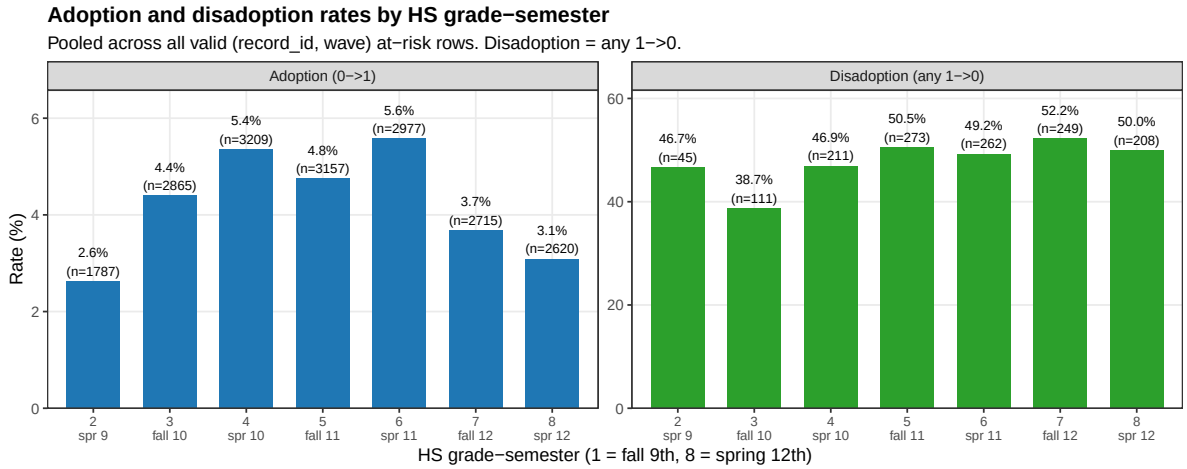



Figure 6: Adoption (left) and disadoption (right) rates by HS grade-semester. Bars labelled with rate and n.

The line graph below puts both trajectories on the same x-axis (dual y-axis: green = disadoption on the left, blue = adoption on the right) so the developmental shape is directly visible:

**Reading.** Adoption follows an inverted-U with a peak around **spring 11th** (gs=6, 5.6%) and the conventional "experimentation peaks mid-HS" pattern — adoption climbs from 2.6% in spring 9th to 5%–5.6% across the gs=4–6 plateau, then falls to 3.1% by spring 12th. Disadoption (any 1→0) drifts gently *upward* from spring 9th (47%) through 12th grade (50%–52%) — the highest semester-level disadoption rate is fall 12th (52.2%). But the spread is narrow: 39%–52% across all seven semesters, summarised by point-biserial correlation $r = +0.048$ (i.e., almost no monotonic trend on the rate). Most variation in both outcomes is across-bin noise rather than a clean grade-semester gradient.

## 11.5 Q-sensitivity: how N students and N events shrink as Q tightens

This subsection shows how (effective N students, N events) for the three disadoption outcomes A / B / C move as Q tightens, and uses that to justify our headline choice $Q = 7$. We span $Q = 4..8$; we omit $Q \in \{9, 10\}$ because **no student in the panel has 9 consecutive observed waves of `past_6mo_use_3`** (a regular four-year HS spans exactly 8 semesters of follow-up, so $\geq 9$ is structurally impossible).

| Q | A (CC: students / events) | B (CC: students / events) | C (CC: students / events) |
|---|---|---|---|
| 4 | 258 / 130 | 300 / 202 | 258 / 33 |
| 5 | 237 / 124 | 271 / 185 | 237 / 29 |
| 6 | 206 / 108 | 233 / 159 | 206 / 21 |
| 7 | 161 / 89 | 182 / 129 | 161 / 15 |
| 8 | 83 / 52 | 91 / 70 | 83 / 7 |

(Counts are after `complete.cases` on the 13-variable PRED set. FULL counts before CC are in `outputs/tables/v4b_table_11_5_Q_sensitivity.csv`.)

**Reading.** Each label below a point is the *one-step* % gain when relaxing Q from the next-stricter value (e.g. the label at Q = 7 is the gain going from Q = 8 to Q = 7). The single biggest one-step jump is at $Q = 8 \rightarrow 7$ across all three outcomes — bolded on the figure:

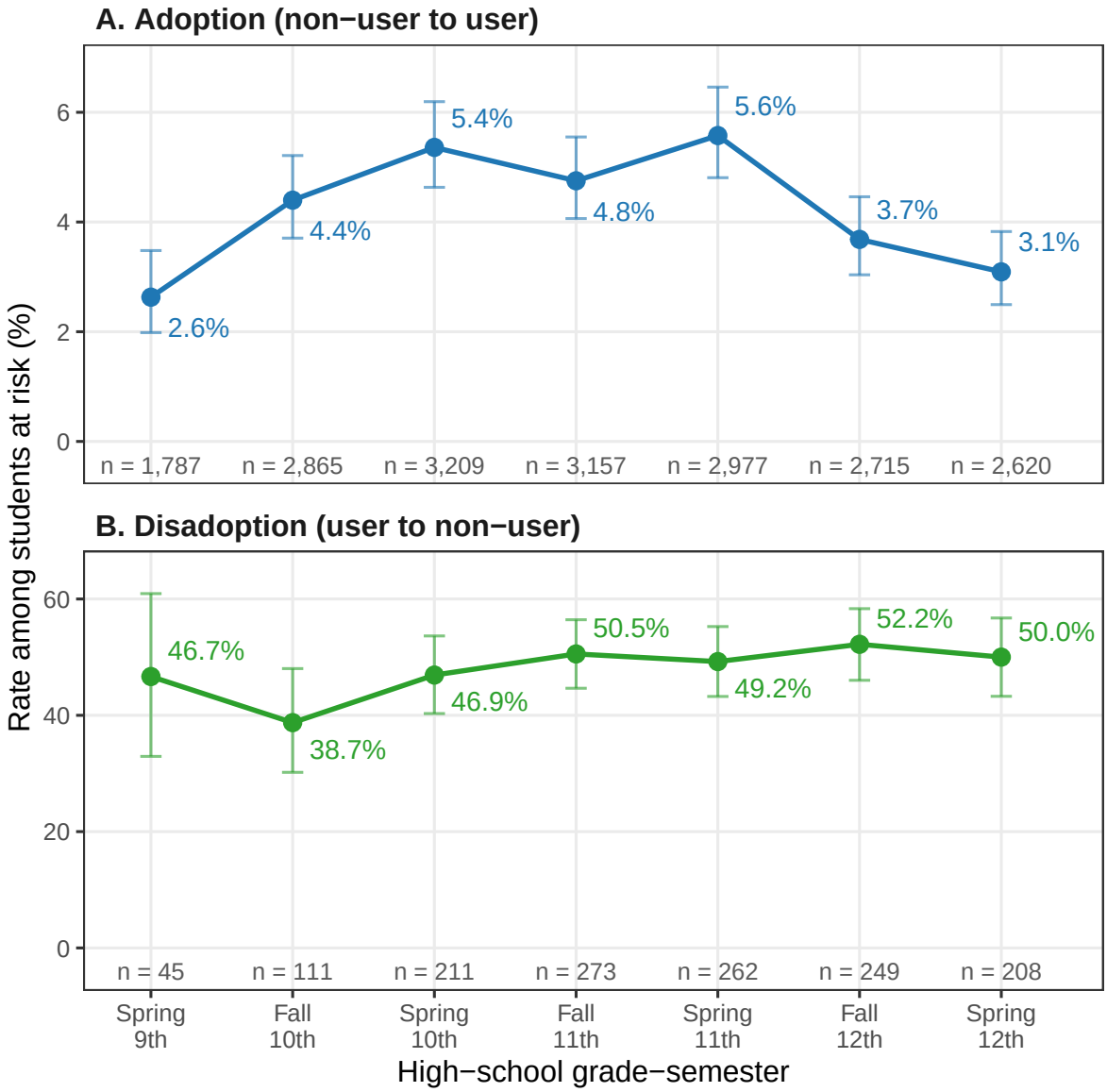



Figure 7: Adoption (top) and disadoption (bottom) rates across the high-school years, each panel on its own y-axis starting at zero; 95% Wilson confidence intervals and number at risk per grade-semester.

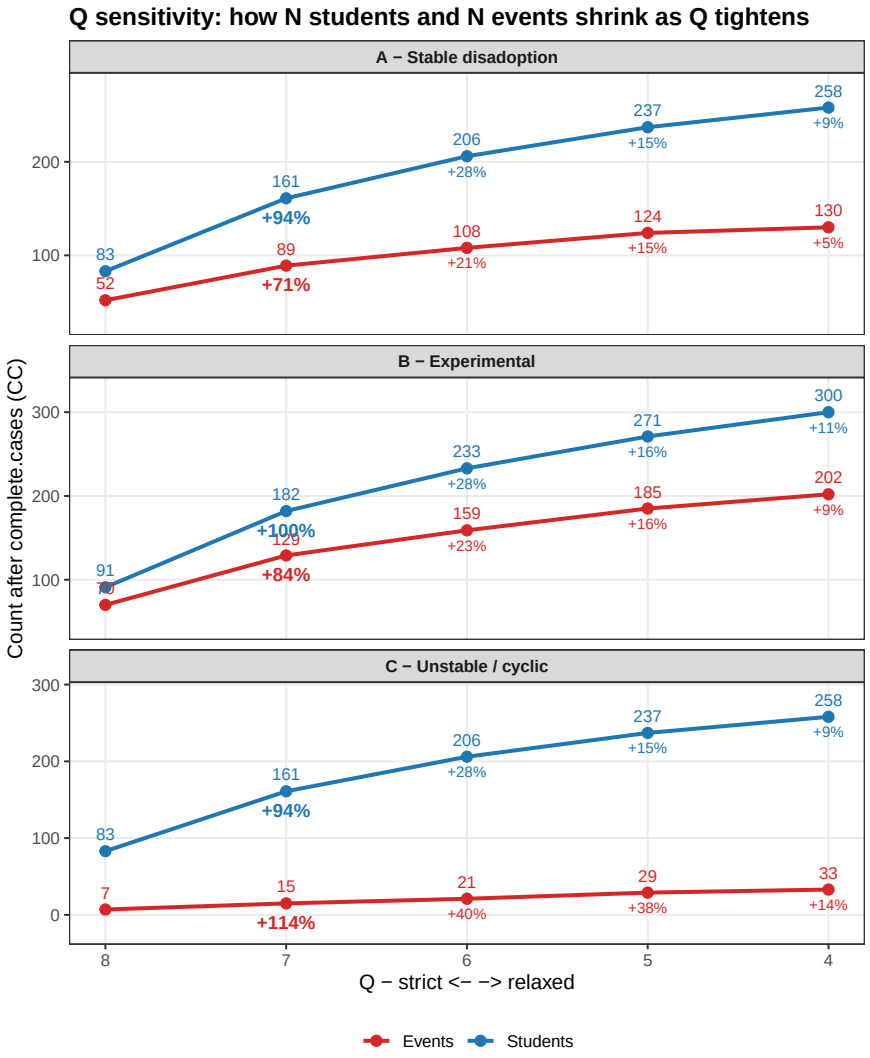



Figure 8: Q-sensitivity per outcome. Three stacked subplots: A (top), B (middle), C (bottom). x-axis runs strict-to-relaxed (Q = 8 on the left). Blue line = students, red line = events. Numbers below each point = % gain vs the immediately stricter Q (per-step relaxation). The bold highlight is **Q = 7**, our recommended sweet spot.

- **A (Stable)**: $Q = 8 \to 7$ adds **+71%** events (52 → 89) and **+94%** students. Subsequent steps add only +14% to +21% per side.
- **B (Experimental)**: $Q = 8 \to 7$ adds **+84%** events (70 → 129) and **+100%** students. Later steps add +14% to +28% per side.
- **C (Unstable)**: $Q = 8 \to 7$ doubles events (**+114%**, 7 → 15). C is too sparse at any Q for further relaxation to buy inferential power.

**Why $Q = 7$ as the headline.** The single biggest jump in usable data per Q step is at $Q = 8 \to 7$ (the bold annotations); every later step trades modest extra power for less observed students. $Q = 7$ keeps 60% of the $Q = 5$ events on A (89 vs 124) and 70% on B (129 vs 185), more than doubles the events from $Q = 8$, and avoids the small-panel separation problem at Q = 8 (the gs_fe block explodes due to event sparsity). Going further ($Q = 6, 5, 4$) buys only marginal additions and blends in students with thin observation histories. $Q = 7$ **is the sweet spot**, and the recommended specification reads off the §6 alt-$E_D$ fits at $Q = 7$ (§6.2).

## 11.6 Network visualization: focal-school dynamics across HS grade-semesters

To visualise the joint adoption / dis-adoption dynamics that §§5–10 fit numerically, we render the friendship network of one focal high school at four half-year snapshots (Spring 9th → Spring 12th grade). The focal school is **School 114** (cohort 2024) — the panel's highest-event school: 115 adoption events ($0 \to 1$) and 81 dis-adoption events ($1 \to 0$) across W2–W8 over 375 students. We restrict the visualisation to the 358 students with at least one friendship tie observed at any wave (17 isolates dropped). Node positions are held fixed across the four panels using two complementary layouts: a force-directed Fruchterman-Reingold layout (Figure 11.6.a) and a community-ordered circular layout (Figure 11.6.b) in which the 9 Louvain communities of the union graph are arranged largest-first around the circle and sorted within community by degree, so intra-community ties become short arcs and inter-community ties cross the centre.

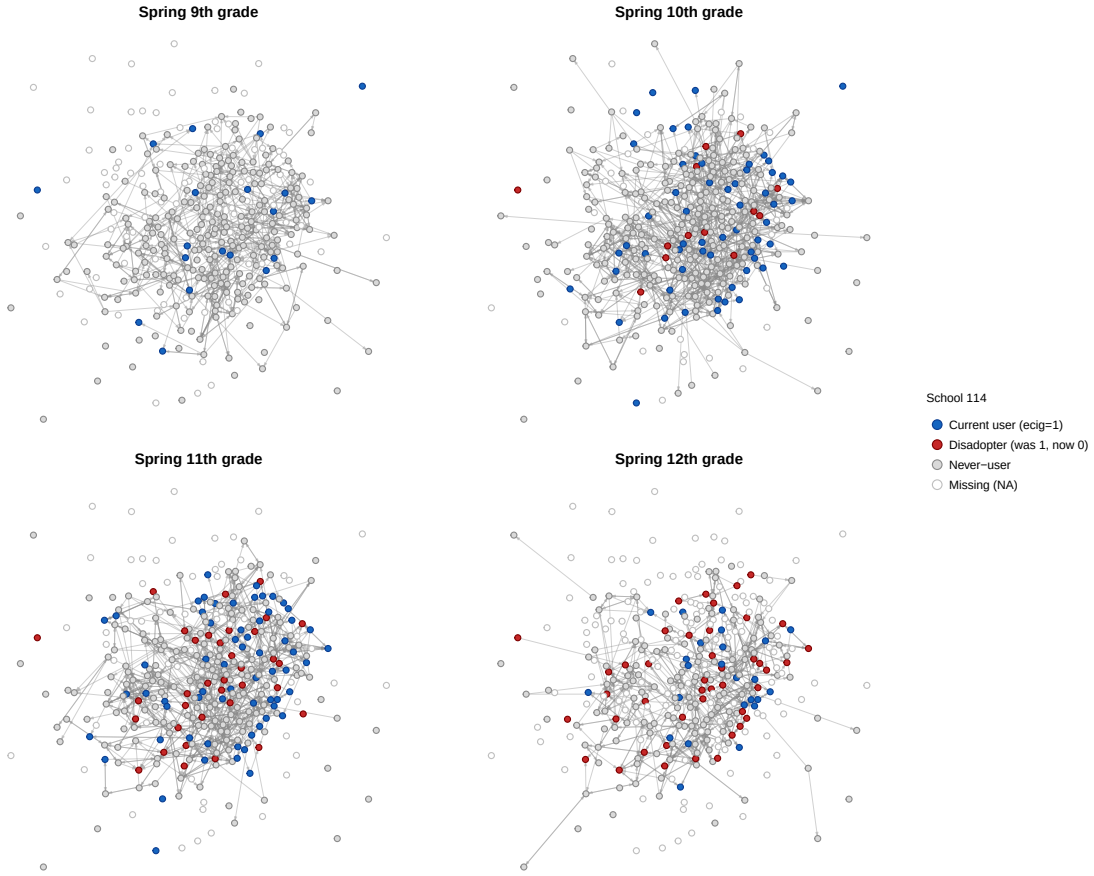



Figure 9: Figure 11.6.a — Friendship-network co-evolution and e-cigarette use, School 114 (cohort 2024), at four half-year snapshots from Spring 9th through Spring 12th grade. Nodes are the 358 students with at least one observed friendship tie across W2–W8 (17 isolates omitted). Node colour encodes each student's e-cigarette status at the displayed wave: blue = current user (past-6-month use); red = dis-adopter (used at a previous wave but not at this one); grey = never-user; white = item-missing. Node positions are held fixed using a Fruchterman-Reingold layout computed once on the union of all observed friendship nominations. The active-user pool (blue) grows from Spring 9th through Spring 11th and then contracts at Spring 12th as dis-adopters (red) accumulate — at Spring 12th, dis-adopters outnumber current users (43 vs 24).

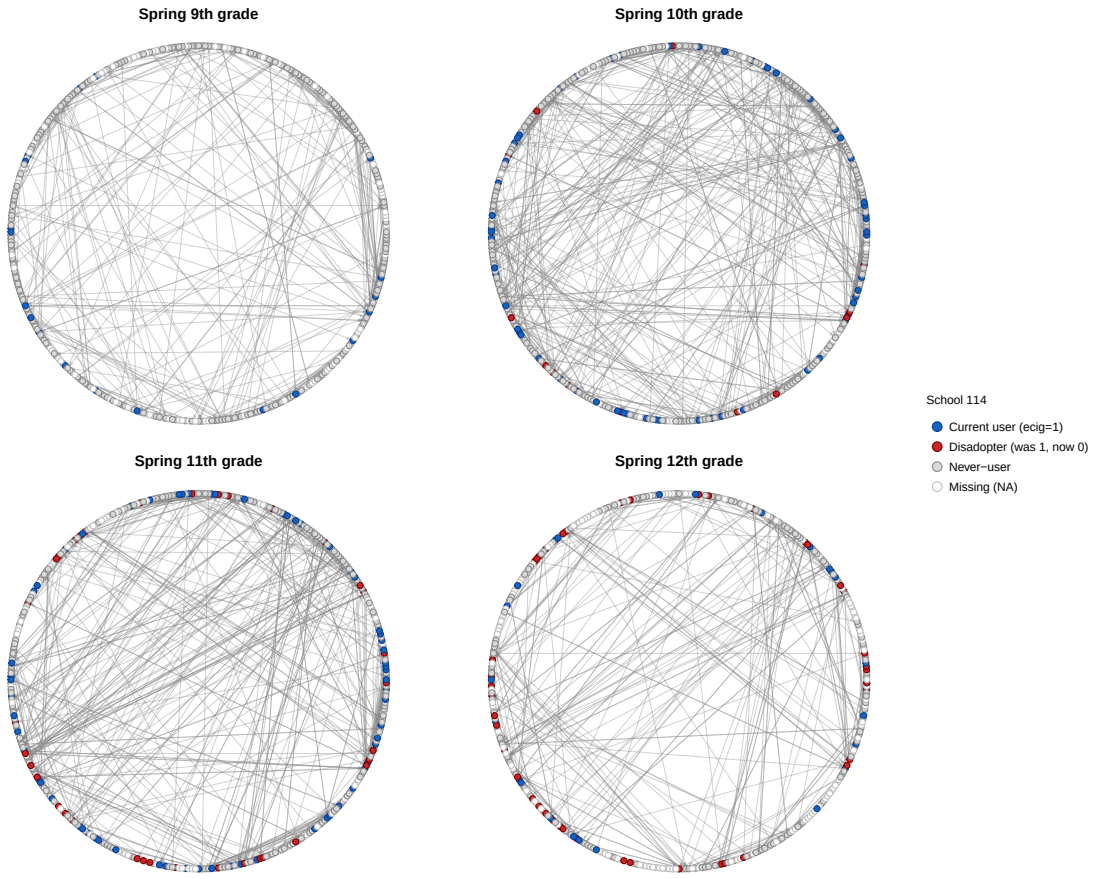



Figure 10: Figure 11.6.b — Same data, colour coding, and time slices as Figure 11.6.a, but with a community-ordered circular layout. Louvain community detection on the union graph yields 9 communities (sizes 62, 54, 50, 48, 44, 29, 27, 25, 19); these are arranged clockwise around the circle, largest first, and within each community nodes are ordered by degree (high → low). The construction puts intra-community ties as short arcs near the perimeter and forces inter-community ties through the centre, making between-community contagion visible at a glance.

All per-wave PDFs (eight standalone wave snapshots and a full 4×2 grid covering every observed wave) are available in the repository at https://github.com/aoliveram/Network-Disadoption/tree/main/outputs/figures.

## Annex — Pipeline

`R/00-config.R` → `R/01b-edges-rebuild.R` (regenerate W1–W10 edges into `data/advance/Cleaned-Data-042326/`) → `R/01-advance-panel.R` (W1–W10 long panel) → `R/02-event-builder.R` (5 modes × 4 Q) → `R/03-network-features.R` (degrees, exposures, $E_D$ variants) → `R/04-regressions.R` (5 families × 4 Q raw fits) → `R/04b-rebuild-tables-OR.R` (re-emit the 20 CSVs as odds ratios).